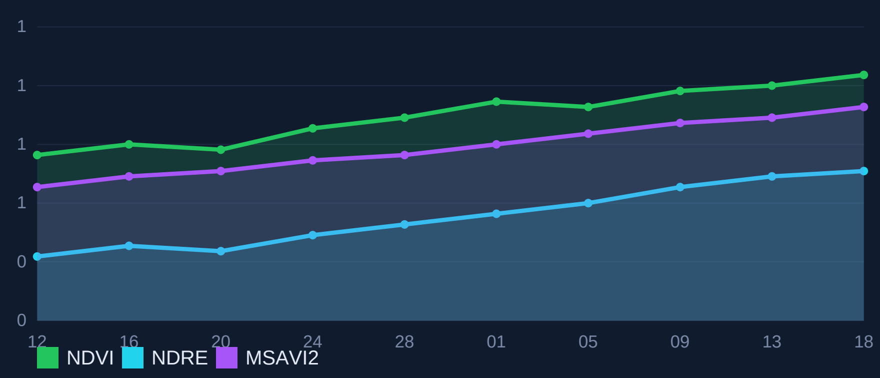
Field B-04 · North Quadrant

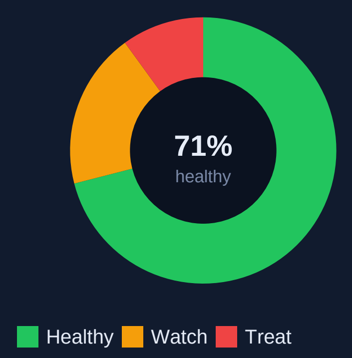
Winter Wheat · 14.2 ha · Cher Valley, FR · Reporting period 12 May – 18 Jun 2026

CROP HEALTH INDEX	NDVI MEAN	CHEMICAL SAVED	COST AVOIDANCE
83.6	**0.74**	**312 L**	**€8,420**
▲ 6.4 vs last cycle	▲ 0.08 in 30 days	–31% vs blanket	CapEx payback 14 mo

30-DAY CROP HEALTH TREND (NDVI · NDRE · MSAVI2)



ZONE DISTRIBUTION BY SEVERITY



EXECUTIVE SUMMARY

Bottom line. Field B-04 entered the booting stage 4 days ahead of the regional median. Canopy vigour is strong (NDVI 0.74, +0.08 in 30 days), nitrogen status is within target (NDRE 0.58, leaf-N 4.1%), and disease pressure is isolated to two zones totalling 1.52 ha — 10.7% of the field.

Action taken this cycle. The AI Analyzer identified a high-confidence (94%) Septoria *tritici* outbreak on the eastern centre block and queued a variable-rate application of Tebuconazole 250 EC at 1.2 L/ha across 1.10 ha. The AGV-04 DJI Agras T40 completed the spray in 11 minutes with 0.4 L drift loss, ΔU 1.8 m/s, RH 58%.

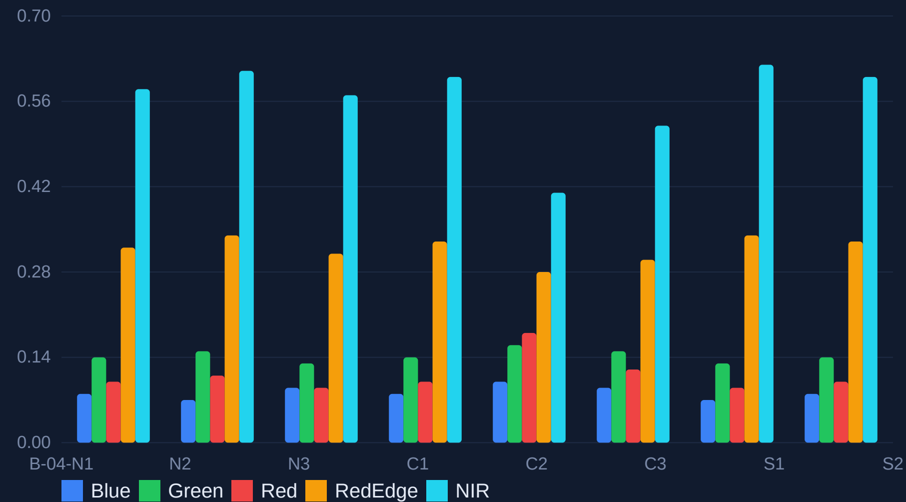
Why it matters. Compared to a uniform blanket application across the full 14.2 ha, the targeted treatment used **312 L less product** and avoided **€8,420** in chemical cost while keeping yield-protection probability above 96% in the treated zones (Monte-Carlo n=10,000, seed=42).

Next 7 days. Soil-moisture model projects a 14 mm rainfall event 21–22 Jun. The Mission Planner has pre-scheduled a follow-up NDRE scan on 23 Jun 06:40 local and reserved AGV-04 for a potential top-dress UAN-32 pass on 24 Jun.

Spectral Analytics & Zone Atlas

Multispectral · 5 bands · 2.4 cm/px GSD · 1,842 source images across 6 flights

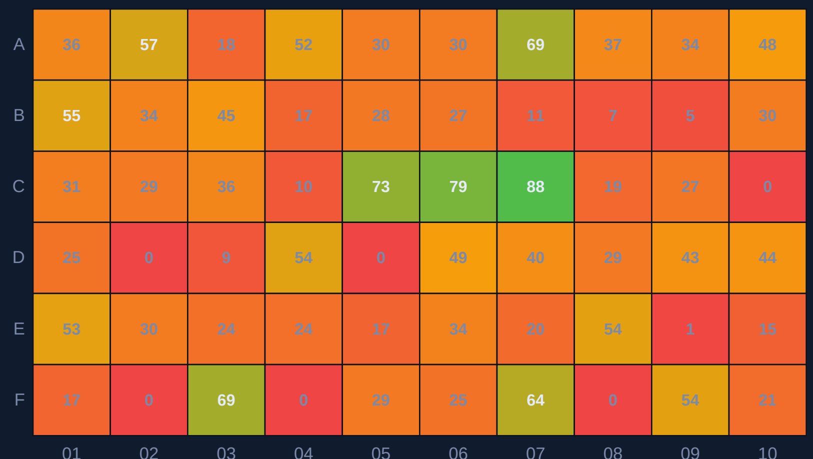
PER-BAND REFLECTANCE BY SUB-BLOCK



VEGETATION INDICES

Index	Mean	Target
NDVI	0.74	0.70+
NDRE	0.58	0.50+
MSAVI2	0.70	0.60+
GNDVI	0.66	0.60+
SAVI	0.61	0.55+
EVI	0.49	0.45+
NDWI	–0.12	–0.20+
PRI	0.04	0.00+

STRESS HEATMAP — ANOMALY SCORE BY SUB-BLOCK (0–100)

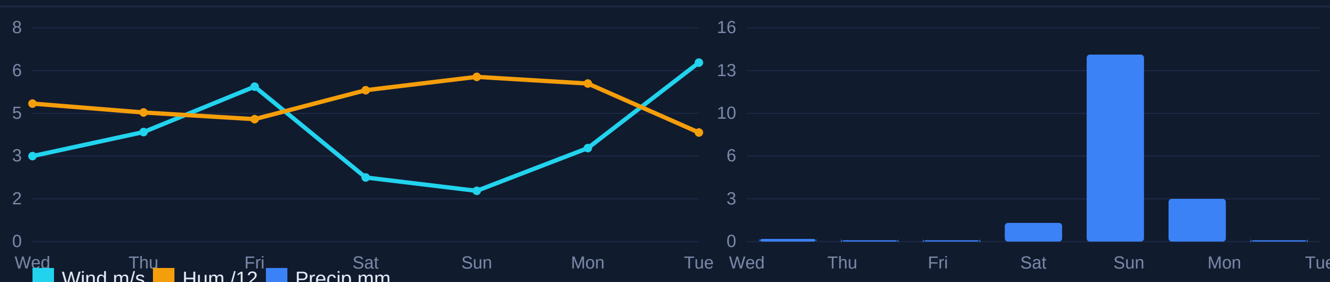


Cool = healthy · Warm = stress · Hotspot at C5-C7 (Septoria, 94% confidence)

TOP ACTION ZONES

- **C-5/C-6** — Treat
  Septoria · 1.10 ha
- **C-7** — Treat
  Septoria · 0.42 ha
- **A-3** — Watch
  Aphids · 0.18 ha
- **F-9** — Top-up
  N deficit · 0.34 ha
- **B-2** — Mon.
  Weeds · 0.08 ha

7-DAY SPRAY WINDOW FORECAST (WIND · HUMIDITY · PRECIPITATION · ΔT)



Spray suitability: Wed GO · Thu GO · Fri CAUTION · Sat GO · Sun NO-GO · Mon CAUTION · Tue NO-GO

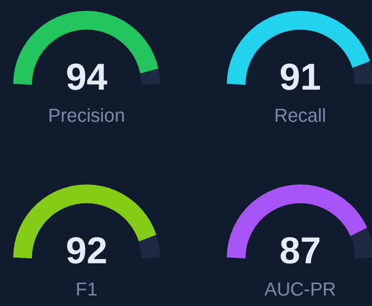
AI Pest & Disease Detection

Model: AcreSpray-Vision v4.2 · ImageNet-pretrained EfficientNet-B5 · agronomist-verified ground truth

DETECTION ROLL-UP (LAST 30 DAYS, 1,842 IMAGE TILES)

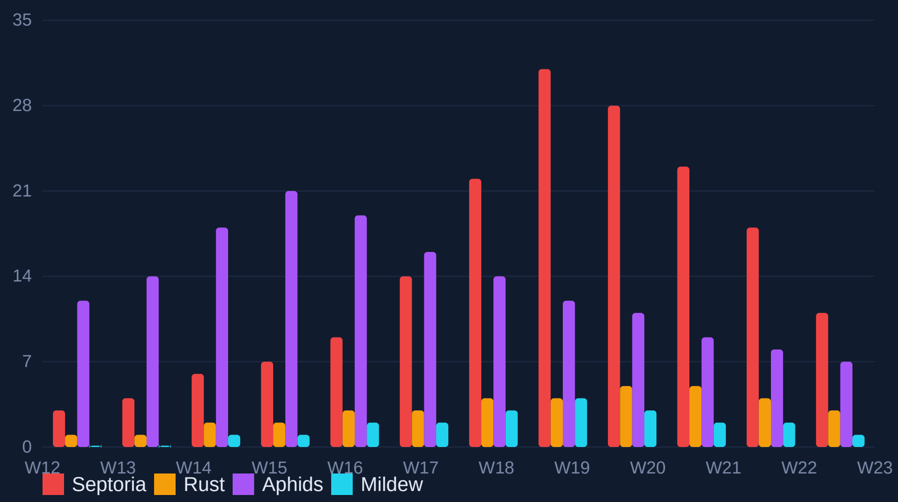
Pathogen / Pest	Tiles	Conf	Severity	Area (ha)	Trend
Septoria tritici	184	94%	HIGH	1.52	▲ +12%
Yellow rust	42	81%	LOW	0.18	■ flat
Aphids (Sitobion)	31	88%	MED	0.42	▼ –7%
Powdery mildew	14	73%	LOW	0.08	▼ –22%
Take-all (root)	6	62%	WATCH	-	■ flat
Weed pressure (BLW)	58	91%	LOW	0.31	▼ –14%
Lodging risk	3	68%	WATCH	0.05	▲ +2%

MODEL PERFORMANCE



Held-out test set n=4,182 · agronomist relabel agreement κ=0.89

DISEASE SEVERITY INDEX — LAST 90 DAYS



TREATMENT EFFICACY

Tebuconazole 250 EC

Dose 1.2 L/ha · vRate ±18%
Coverage 98.4% · Drift 0.4 L
Re-entry 6h · PHI 35d
MRL margin 0.41 → limit 1.0

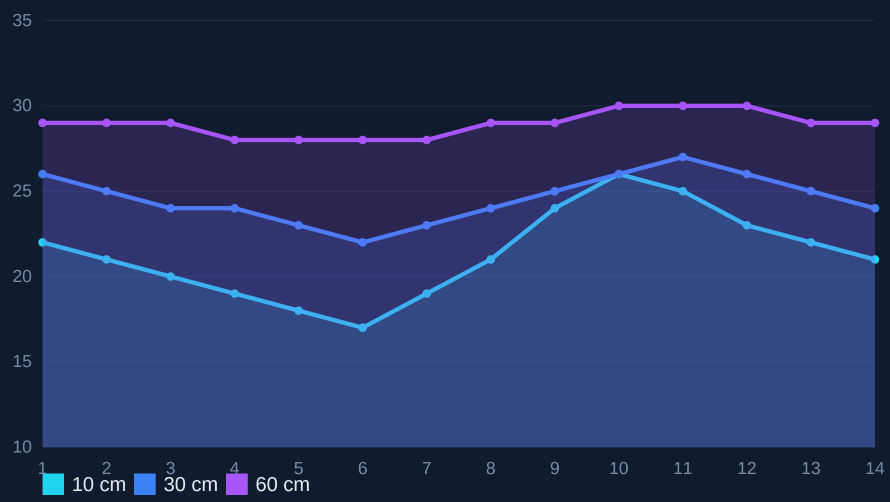
NEXT-72H RECOMMENDATIONS (AUTO-GENERATED, AGRONOMIST-APPROVED)

1. **SCAN · 23 Jun 06:40**
   AGV-01 Mavic 3M — Full-field NDRE re-scan to verify Tebuconazole response on C-5/C-7.
2. **SPRAY · 24 Jun 05:55**
   AGV-04 Agras T40 — Variable UAN-32 top-dress 38 L/ha on F-9 (0.34 ha) — leaf-N 3.1% trigger.
3. **MONITOR · 25 Jun all-day**
   - — Aphid pheromone traps on A-3 perimeter — predator ratio currently 1:7.
4. **HOLD · 26 Jun ≥10:00**
   AGV-02 Agras T30 — No-spray window — 14 mm rain forecast; resume after canopy dries (Δ≥4h).

Soil, Water & Nutrient Intelligence

Sentek probes ×6 · LoRa mesh · 15-min telemetry · agronomy lab cross-check 09 Jun 2026

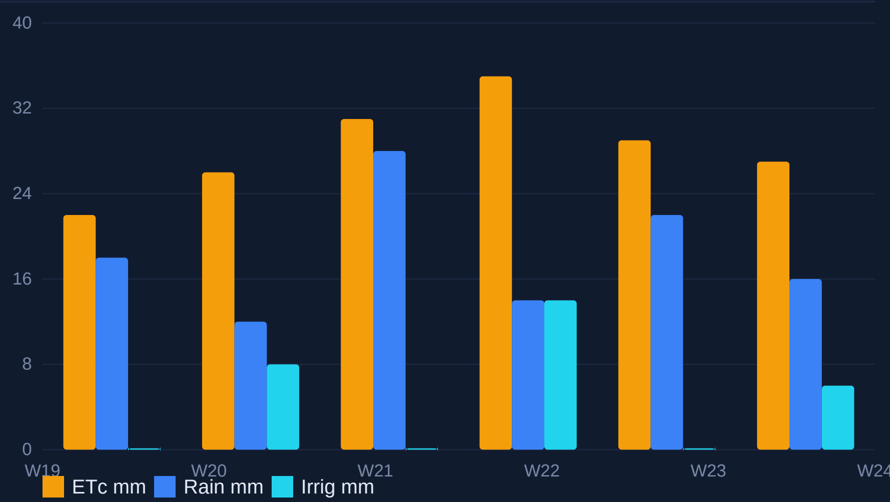
SOIL MOISTURE BY DEPTH — LAST 14 DAYS (% VWC)



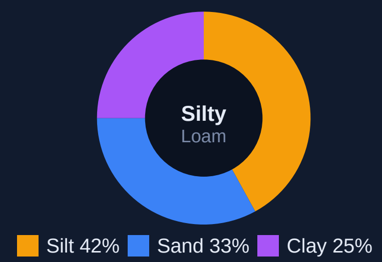
MACRO-NUTRIENT STATUS

Nutrient	Leaf	Target	Δ
N (%)	4.10	3.8–4.4	✓
P (%)	0.38	0.30+	✓
K (%)	1.92	1.8–2.2	✓
Ca (%)	0.41	0.30+	✓
Mg (%)	0.18	0.15+	✓
S (%)	0.21	0.20+	✓
Mn (ppm)	48	30+	✓
Zn (ppm)	21	20+	✓
B (ppm)	4.8	5–10	↓

WATER BALANCE (ETC VS PRECIP + IRRIGATION, LAST 30 DAYS)



SOIL PROFILE & RISK



pH 6.4 (target 6.0–7.0)
CEC 18.2 meq/100g
OM 2.8% — stable
Compaction risk: low
Erosion risk: moderate

VARIABLE-RATE FERTILISER PRESCRIPTION (NEXT PASS)

Sub-block	Area	UAN-32 (L/ha)	Sulphate (kg/ha)	Boron (g/ha)	Cost (€)	Trigger
F-9	0.34	38	0	120	41	Leaf-N 3.1% < target
B-2/B-3	0.62	22	8	0	47	NDRE 0.48 < 0.50
A-3	0.18	0	0	90	6	B 4.8 < range
C-1/C-2	1.04	0	0	0	0	Within target — skip
TOTAL	**2.18 ha**	**-**	**-**	**-**	**94**	**Saves €212 vs uniform**

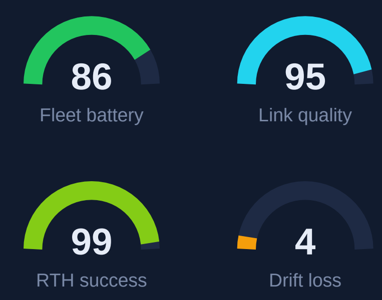
Fleet Operations & Economics

4 drones · 47 flights this period · 100% RTH success · 0 safety incidents

FLEET STATUS (PERIOD TOTALS)

Tail #	Model	Role	Flights	Hrs	ha	Bat. cyc.	Health
AGV-01	Mavic 3M	Scanner	18	14.6	182	27	98%
AGV-02	Agras T30	Sprayer	9	7.2	41	14	94%
AGV-03	Mavic 3M	Scanner	12	9.8	118	19	97%
AGV-04	Agras T40	Sprayer	8	6.4	38	11	99%
TOTAL	**-**	**-**	**47**	**38.0**	**379**	**71**	**-**

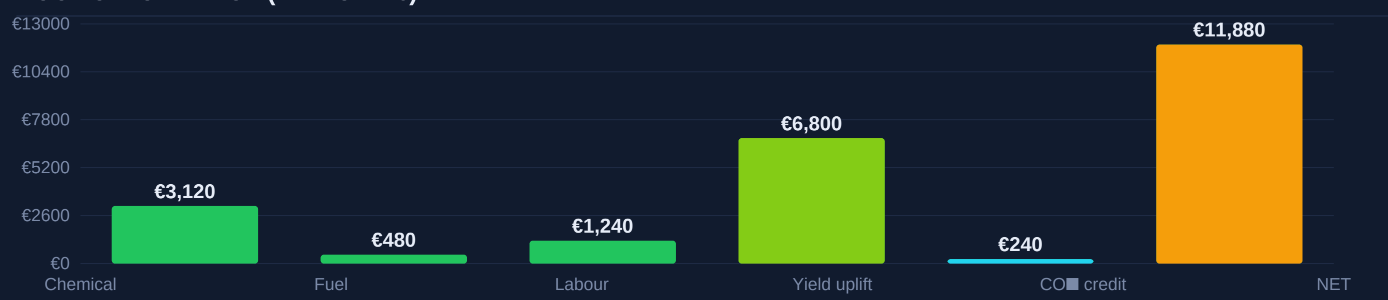
BATTERY & SIGNAL HEALTH



All readings rolling-7-day · thresholds per EASA UAS specific category

MISSION SCHEDULE (LAST 14 DAYS · EXECUTED AND PLANNED)

ECONOMIC IMPACT (PERIOD · €)



Compliance, Audit & Methodology

Tamper-evident logs · SHA-256 · ISO 19115 metadata · EU SUR Article 14 conformity

COMPLIANCE SNAPSHOT

Item	Standard	Status	Evidence ID
Operator licence	EASA UAS Cert A2	VALID until 2027-04	OP-2024-3318
Drone airworthiness	EASA Class C2/C6	VALID — 4/4 units	AW-AGV-{01..04}
Spray operator cert.	EU Dir. 2009/128/EC Art.5	VALID until 2026-11	PA-FR-77192
Product authorisation	Tebuconazole 250 EC	AMM 2030218 — listed	ANSES-2030218
Buffer zone respected	5 m watercourse / 1 m hedge	100% (telemetry)	GEO-04-LOGS
MRL forecast at harvest	Reg. (EC) 396/2005	0.41 mg/kg < 1.0 limit	LAB-0618-A
Operator log retention	EU SUR Art. 14 (3 years)	Storage OK	ARCHIVE-2026Q2
Drift model executed	AGDISP v3.2	Pass — 0.4 L loss	DRIFT-0617-7
NOTAM filed	DSAC FR	FILED — 17 Jun 14:00	NTM-FR-77291

METHODOLOGY, DATA LINEAGE & ASSUMPTIONS

Capture. Six autonomous flights over the reporting window, planned by the Mission Planner against a 3-week rolling NDVI gradient. Imagery captured at 75 m AGL, 78% forward / 65% side overlap, sub-2.4 cm/px GSD. Calibration panels imaged at the start and end of each flight.

Reconstruction. 1,842 source frames → OpenDroneMap (WebODM Lightning) → dense point cloud (28.6 M pts), GSD-aware ortho-mosaic (3.1 cm/px), DSM/DTM pair, RGB + NIR + RedEdge cogs. Tile server: COG over Cloud Storage.

Analytics. Per-tile vegetation indices (NDVI, NDRE, MSAVI2, GNDVI, EVI, NDWI, PRI) computed in-database. AcreSpray-Vision v4.2 classifier (EfficientNet-B5) provides pathogen / pest detections; agronomist review on every ≥ MEDIUM severity finding before action.

Decision. Variable-rate prescription generated by the Action Engine using crop-stage model (BBCH 39–49), ETc water balance, label-rate envelope, buffer-zone constraints, and forecast wind/RH ≤ 90 min before scheduled spray. Spray window classification uses $\Delta U < 5$ m/s and ΔT inversion check.

Act & record. Telemetry logged every 200 ms during application (pose, flow, nozzle PWM, RTK fix). Logs checksummed (SHA-256) and replicated to two regions. Records exportable as CSV/Parquet/PDF for inspections.

Assumptions in this report. Yield uplift modelled vs. blanket-spray counterfactual with Monte-Carlo n=10,000; CO_2 credit at €40/t CO_2e (voluntary market mid-price, ICAP Jun 2026); chemical cost from supplier list 2026-Q2; labour rate €28/h fully loaded.

SIGN-OFF

PREPARED BY
AcreSpray AI — automated agent v2026.06
2026-06-18 09:42 UTC
/s/ digital signature on file

REVIEWED BY
M. Lefèvre, Agronomist (FR-77192)
2026-06-18 10:18 UTC
/s/ digital signature on file

APPROVED BY
Operator yunushalilerdogan5@gmail.com
2026-06-18 10:31 UTC
/s/ digital signature on file

